Mamba learning rate and inner width

Exact original GraphCast loss | 1 degree | batch 4 | di16 and di32 | 10-day forecasts

Completed slower, joint and faster Mamba comparisons

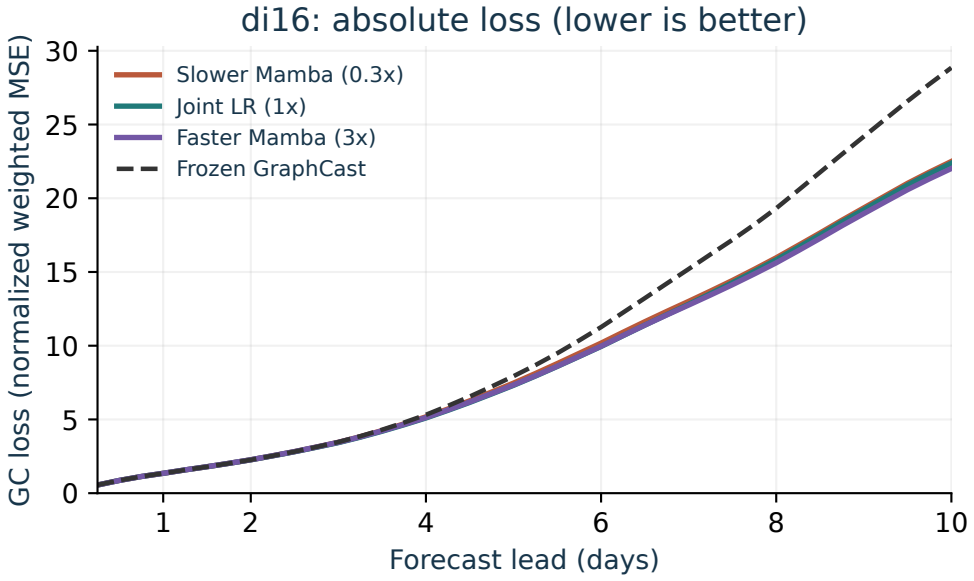
Inner width	Mamba LR	Final rollout	SWA rollout	SWA day 10	SWA vs joint
di16	Slower Mamba (0.3x)	13.63%	14.70%	21.98%	-0.97 pp
di16	Joint LR (1x)	15.26%	15.67%	22.44%	+0.00 pp
di16	Faster Mamba (3x)	15.26%	16.34%	23.68%	+0.67 pp
di32	Slower Mamba (0.3x)	15.21%	15.82%	22.96%	-0.48 pp
di32	Joint LR (1x)	15.69%	16.29%	23.32%	+0.00 pp
di32	Faster Mamba (3x)	15.14%	15.73%	22.75%	-0.56 pp

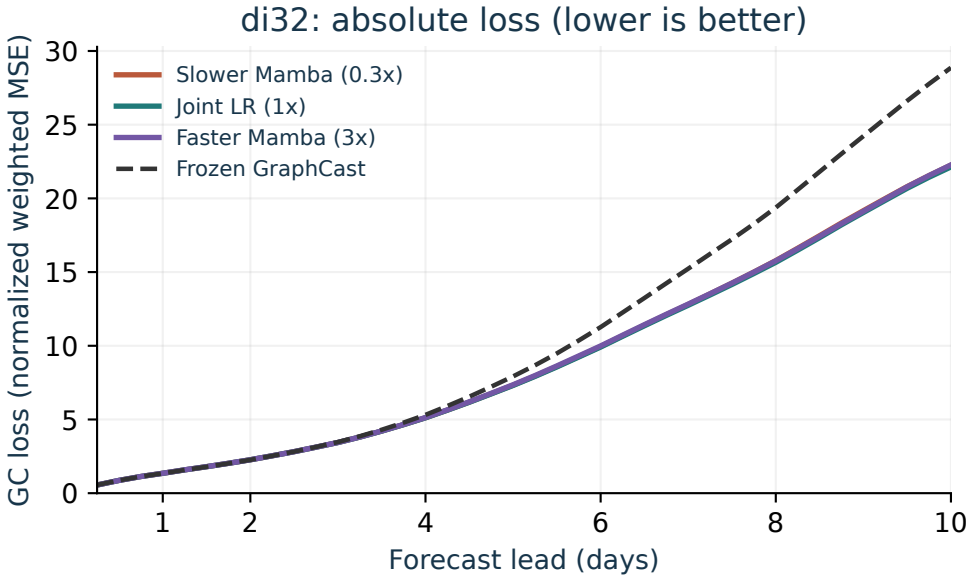
di16: slower Mamba 14.70% vs joint 15.67% rollout reduction (-0.97 percentage points).
di32: slower Mamba 15.82% vs joint 16.29% rollout reduction (-0.48 percentage points).
Increasing di16 to di32: joint +0.62 pp; slower Mamba +1.11 pp (SWA).
Different warm-up weights and one training seed limit causal conclusions.

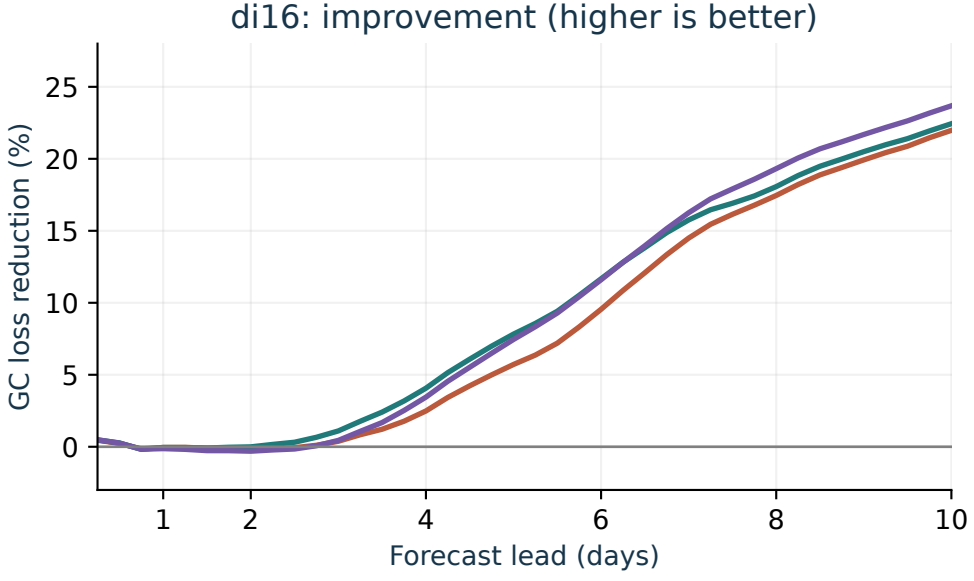
Higher reduction is better. All headline scores require 32 completed forecast starts.
Rollout: aggregate all 40 six-hour losses before taking the ratio. Day 10: the final lead only.
Final = branch update 2,000. SWA = average of branch checkpoints 500, 1,000, 1,500, 2,000.

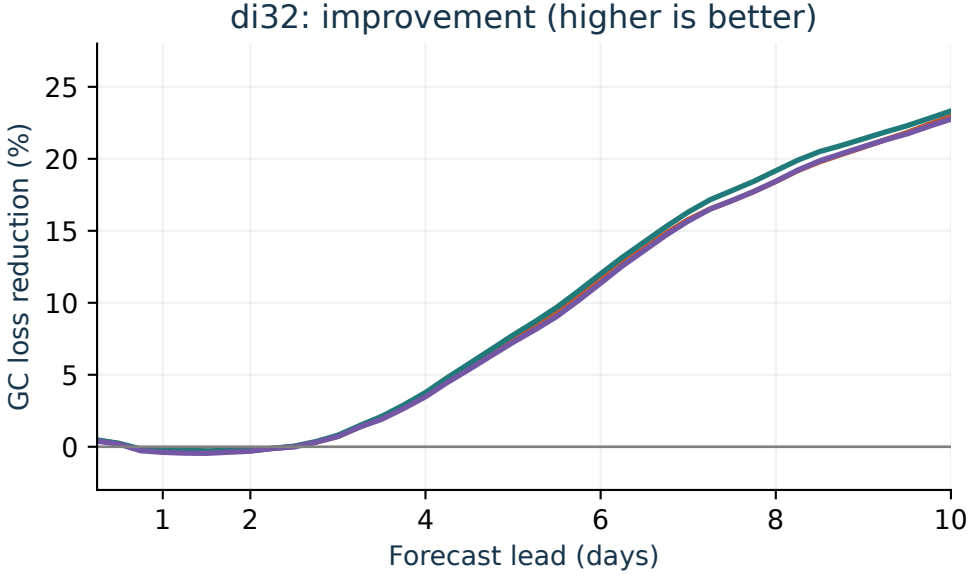
Forecast loss across all 10 days

SWA checkpoints | complete 32-start evaluations | solid colors identify Mamba learning-rate multipliers





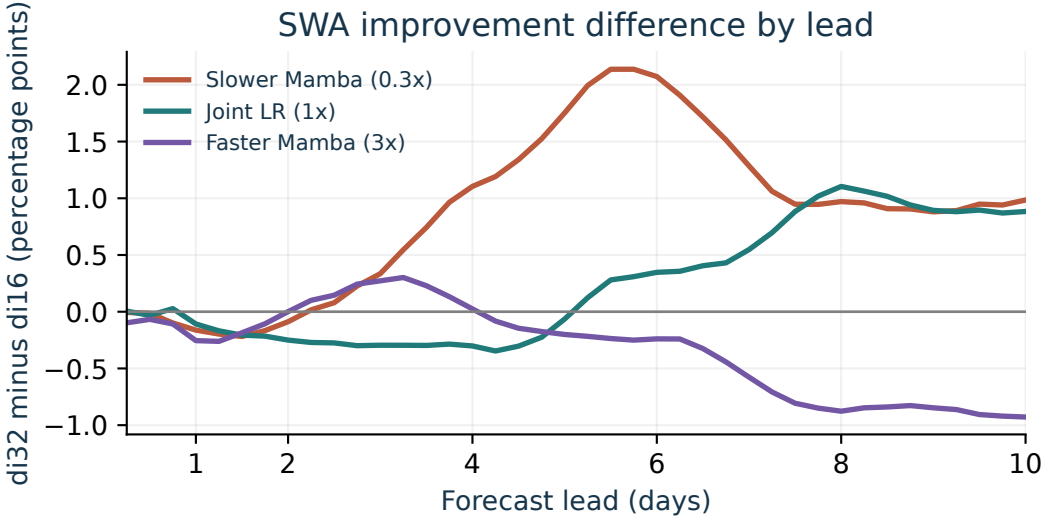




Each improvement uses its own matched baseline.

How the result depends on di

di is the Mamba inner dimension; spatial width stays 512 and BC groups stay 1



SWA	Day 1	Day 5	Day 10
di16 slow	-0.05%	5.72%	21.98%
di16 joint	-0.05%	7.82%	22.44%
di16 fast	-0.13%	7.45%	23.68%
di32 slow	-0.21%	7.47%	22.96%
di32 joint	-0.16%	7.76%	23.32%
di32 fast	-0.39%	7.25%	22.75%

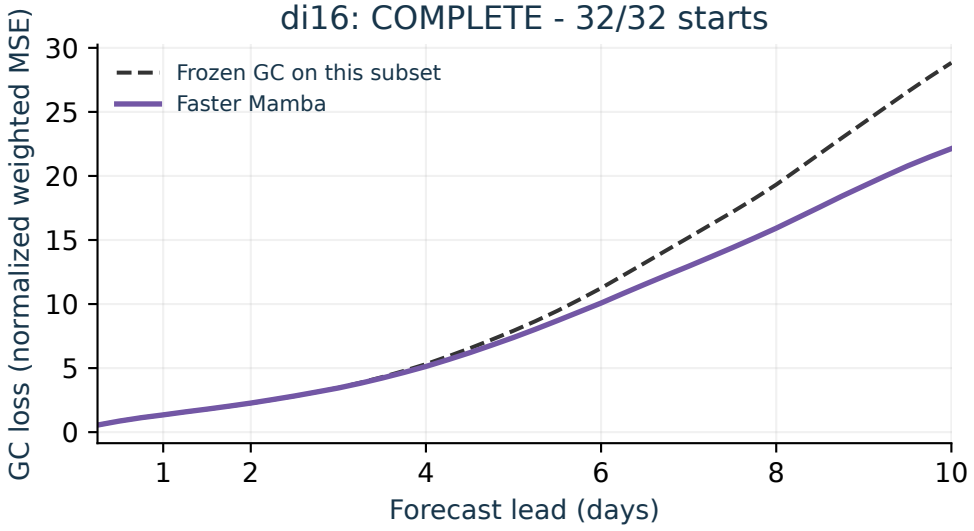
Arm	Own warm-up	SWA rollout	Gain over own warm-up
di16_joint	9.06%	15.67%	+6.61 pp
di16_slow_mamba	7.13%	14.70%	+7.58 pp
di32_joint	7.37%	16.29%	+8.92 pp
di32_slow_mamba	7.70%	15.82%	+8.12 pp

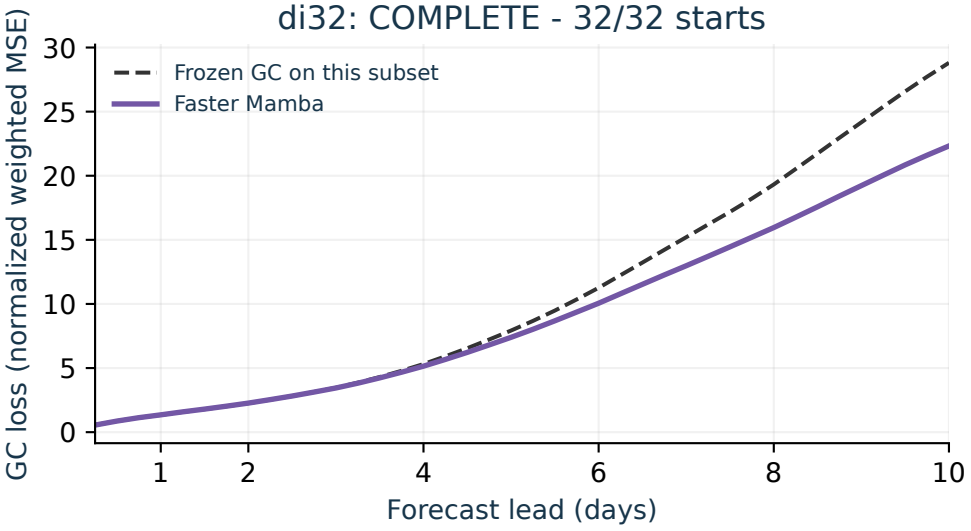
The slower-Mamba rollout score increases more with di (+1.11 pp) than the joint score (+0.62 pp).
The gap to joint narrows from 0.97 pp at di16 to 0.48 pp at di32.
These are descriptive differences: all ten warm-up parameter hashes differ. In particular, di16 slower Mamba starts weaker but gains more during continuation. A shared-checkpoint comparison is needed to attribute the final-score difference to learning rate alone.

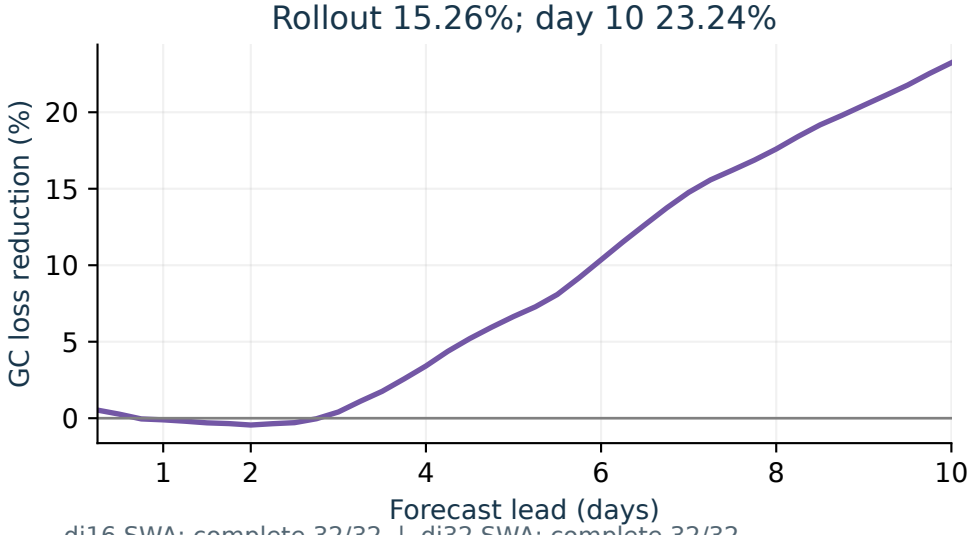
Faster Mamba: evaluation progress

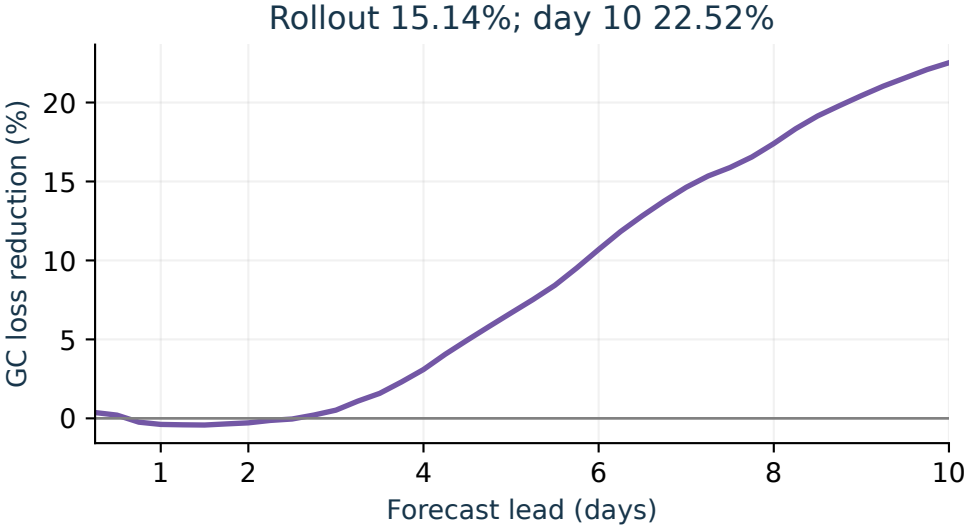
Peak Mamba LR 3e-4; spatial LR 1e-4 | final branch checkpoint shown here; SWA reported separately

Both final checkpoints have completed the matched 32-start evaluation.







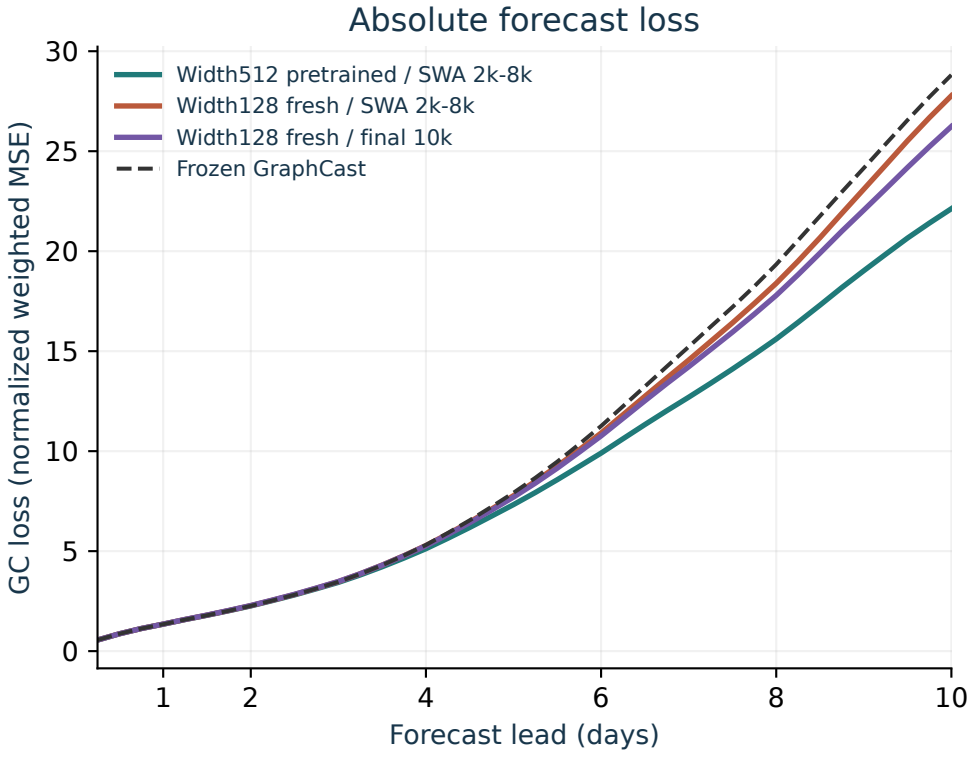


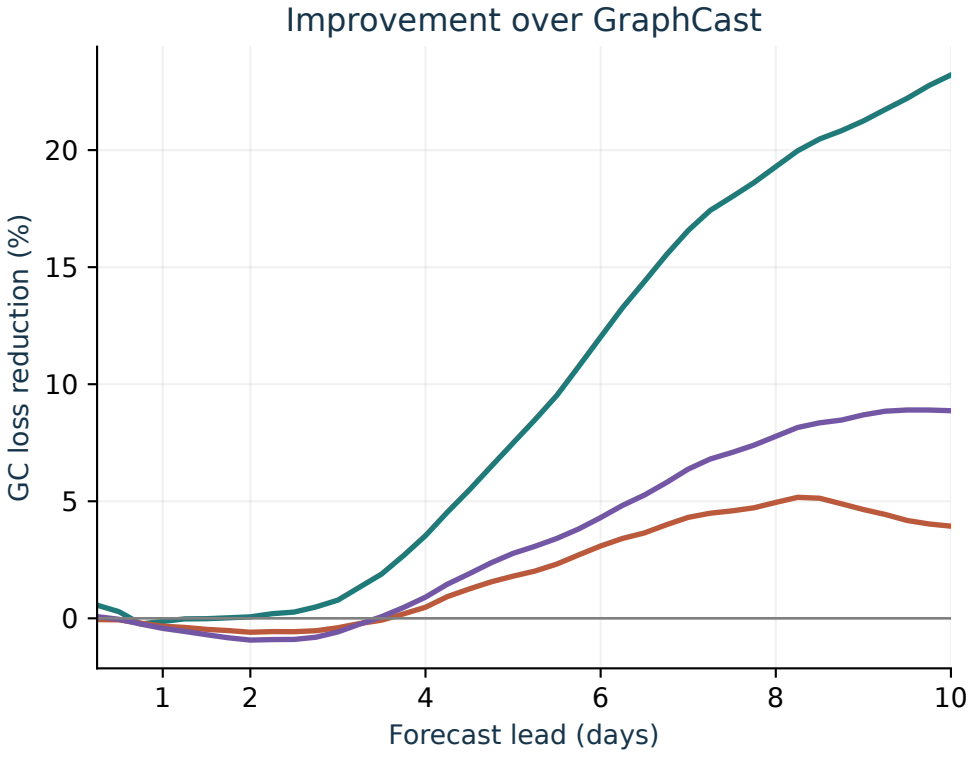
di16 SWA: complete 32/32 | di32 SWA: complete 32/32

Snapshots retain the exact input bytes and completed start IDs. Missing per-start losses prevent a matched-subset comparison.

Smaller residual GraphCast: cost and skill

Separate batch-1 experiment | 10,000 updates | di16 / BCG1 | same 32 starts and 40 forecast leads





Residual branch / artifact	Parameters	Rollout reduction	Day-10 reduction
Width512 pretrained / SWA 2k-8k	10,348,879	16.28%	23.21%
Width128 fresh / SWA 2k-8k	730,447	3.63%	3.94%
Width128 fresh / final 10k	730,447	6.33%	8.87%

Width128 has 92.94% fewer residual parameters, retaining 22.28% of the reference SWA benefit.
Width and pretrained-versus-fresh initialization both change; this does not isolate width. Baseline width stays 512.

All arms, methods and evidence

Completed-only score table | exact normalized, latitude-, pressure- and variable-weighted GraphCast MSE

Arm	Warm-up rollout	Final rollout	SWA rollout	SWA day 10
di16_joint	9.06%	15.26%	15.67%	22.44%
di16_slow_mamba	7.13%	13.63%	14.70%	21.98%
di16_frozen_spatial	9.24%	12.15%	11.59%	17.34%
di32_joint	7.37%	15.69%	16.29%	23.32%
di32_slow_mamba	7.70%	15.21%	15.82%	22.96%
di32_frozen_spatial	7.32%	12.02%	11.88%	17.85%
di16_fast_mamba	9.45%	15.26%	16.34%	23.68%
di16_slow_spatial	9.03%	15.71%	15.23%	22.08%
di32_fast_mamba	7.94%	15.14%	15.73%	22.75%
di32_slow_spatial	6.24%	15.83%	15.02%	21.99%

METHOD
Loss reduction = 100 x (1 - model loss / matched frozen-baseline loss). Rollout losses aggregate
all starts and 40 six-hour leads before the ratio; percentages are never averaged across leads.
Training: 500 private warm-up + 2,000 branch updates; batch 4; seed 22; Mamba1; residual width512/MP2;
di16 or di32; BCG1; state size16; 2 temporal layers. Peak LR 1e-4 with Mamba factors 0.3 / 1 / 3,
50-update warm-up, cosine to 1e-5 before group scaling; AdamW (0.9, 0.98), weight decay 1e-4.
Evaluation: 2022, same 32 start IDs, cold full feedback, zero state, no history warm start.
Checked completeness, IDs, metadata, loss ratios and arrays. Maximum matched baseline deviation: 0.556%.
Partial previews have fewer starts; no common per-start losses are saved for retrospective matching.

EVIDENCE AND REPRODUCTION
Report inputs, hashes and checks: v24_mamba_lr_di_10day_2026-09-14.sources.json
Numerical summary: v24_mamba_lr_di_10day_2026-09-14.csv
Input snapshots and per-lead CSV: plots/analyze_models/data/resolution_eval/v24_mamba_lr_di_10day_2026-09-14/
Generator: scripts/analyze_models/report_v24_mamba_lr_di.py (GraphCast environment).
One seed, differing warm-up weights and a development evaluation set: no significance claim.